James Lin

Mechanical engineering · Downingtown STEM Academy, class of 2027

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LinkedIn · Instagram · X
Chester County, PA

Education

Downingtown STEM Academy — Class of 2027

IB Diploma Programme · HL Physics, Design Technology, Math AA, Economics

GPA 3.93 (unweighted) · SAT 1550 (790 Math)

Experience

Engineering Intern, Nth Solutions — 2026 to present

Coatesville, PA

- Work on vibration-analysis software using FFTs and signal processing; taught myself the FFT math from Gilbert Strang's MIT OpenCourseWare lectures
- Designed CAD jigs, wrote an analysis script comparing FFT and OLS methods, and optimized VKF bandwidth
- Explain the analysis software to new interns

High School Intern, NASA SEES — 2025

UT Austin Center for Space Research

- Selected as 1 of about 80 students from more than 3,000 applicants nationwide
- CAD lead for a proof-of-concept Titan airship: designed the gondola, rear vectoring propeller, and skis in Fusion 360, and modeled flow in Earth and Titan conditions with Autodesk CFD
- Presented our research at the AGU Fall Meeting in New Orleans
- Co-wrote a microgravity air filtration proposal that was runner-up out of about 100, making us the alternate team

Content Creator — Ongoing

@jameslarpsandmakescoolstuff on Instagram · @JamesMakes29 on X

- Document my engineering builds for 10,000+ followers across platforms, with 4M+ total views
- Followed by Jeff Bezos on X
- Sponsored by Bambu Lab, LDO Motors, and Cytron Technologies for parts and equipment

Front Desk, Server, Busser, Zwei Restaurant — 2019 to 2026

Exton, PA · family restaurant, about 20 hours a week

- Took orders, ran the front desk, and bussed tables in a fast-paced restaurant
- 3D printed fixes for the restaurant and handled small repairs and printer issues

Awards

- **FTC World Championship**, Alliance Captain (Lovelace Division) and Sustain Award, team 7244 Out of the Box Robotics; drive team and CAD
- **TSA Nationals**, 3rd of 187 in Geospatial Technology
- **TSA States**, 3rd of 70+ in CAD Engineering
- **Delaware Valley Science Fairs**, 3rd of about 50 team projects and West Pharmaceuticals Best Team Award ($700), nitinol robotic claw

Projects

- **Differential swerve drive**, 8 NEMA17 steppers on an ESP32, driven from a phone over WiFi
- **Robot arm and nitinol claw**, gear-driven stepper arm with a shape-memory-alloy compliant claw
- **Mechanical integrator**, disk-and-wheel integrator with a belt-driven differential, plus a math write-up
- **Eye-operated switch**, IB Design Technology IA for students with multiple disabilities (in progress)
- **Ultralight drone**, generatively designed arms checked with FEA, tuned in Betaflight

Leadership and service

- **Class Council**, Vice President, 11th and 12th grade
- **stemHACKs**, Vice President; my CAD workshops helped 3 students qualify for TSA States
- **Campfire Chester County Hackathon**, organizer of a 6-hour game jam with 25+ participants
- **Chester County Library Makerspace**, weekly volunteer; teach CAD, slicing, and 3D printing, maintain the Ender 5 and Voron 2.4, and built a machine that turns plastic bottles into filament
- **National Honor Society**, member since 11th grade

Skills

CAD and simulation	Fusion 360 (incl. generative design), Onshape, SolidWorks, Autodesk FEA and CFD
Fabrication	FDM 3D printing (Bambu, Voron, Ender), printer building and maintenance
Electronics and code	ESP32, Arduino, stepper drivers, Betaflight, Python, Java, Swift
Analysis	FFT and signal processing for vibration data